

tf.parallel_stack Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/parallel_stack

Stacks a list of rank- R tensors into one rank- $(R+1)$ tensor in parallel.

Function Signatures

$(R+1)$

```
tf.parallel_stack( values, name='parallel_stack' )
```

Code Examples

```
tf.parallel_stack(  
  values, name='parallel_stack'  
)
```

```
tf.parallel_stack(  
  values, name='parallel_stack'  
)
```

```
x = tf.constant([1, 4])  
y = tf.constant([2, 5])  
z = tf.constant([3, 6])  
tf.parallel_stack([x, y, z]) # [[1, 4], [2, 5], [3, 6]]
```

```
x = tf.constant([1, 4])  
y = tf.constant([2, 5])  
z = tf.constant([3, 6])  
tf.parallel_stack([x, y, z]) # [[1, 4], [2, 5], [3, 6]]
```

```
tf.parallel_stack([x, y, z]) = np.asarray([x, y, z])
```

```
tf.parallel_stack([x, y, z]) = np.asarray([x, y, z])
```

Documentation

Stacks a list of rank-R tensors into one rank-(R+1) tensor in parallel.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.parallel_stack`

Requires that the shape of inputs be known at graph construction time.

Packs the list of tensors in values into a tensor with rank one higher than each tensor in values, by packing them along the first dimension.

Given a list of length N of tensors of shape (A, B, C); the output tensor will have the shape (N, A, B, C).

For example:

The difference between `stack` and `parallel_stack` is that `stack` requires all the inputs be computed before the operation will begin but doesn't require that the input shapes be known during graph construction.

`parallel_stack` will copy pieces of the input into the output as they become available, in some situations this can provide a performance benefit.

Unlike `stack`, `parallel_stack` does NOT support backpropagation.

This is the opposite of `unstack`. The numpy equivalent is

Returns

Raises

eager compatibility

`parallel_stack` is not compatible with eager execution.